Review Article

Beyond Adhesives: The Multidimensional Role of Colophony in Industry and Medicine

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ABSTRACT

Colophony, commonly known as rosin, is a natural resin derived primarily from coniferous trees like pines. It is composed predominantly of resin acids, particularly abietic acid, along with a neutral fraction. This complex mixture serves as a versatile industrial raw material, with applications in adhesives, cosmetics, paints, inks, soldering fluxes, and medical products. Its eco-friendly properties, renewability, and affordability have earned it the designation green oil. Despite its extensive usage, colophony poses potential health risks, including allergic contact dermatitis (ACD) and occupational asthma, primarily due to its oxidized resin acids. Diagnosis and prevention of these adverse effects remain challenging. Modified forms of colophony, while reducing some allergenic components, may introduce new sensitizers like maleopimaric acid. Colophony also demonstrates significant pharmacological activities. Its derivatives exhibit wound healing, cytotoxicity, antimicrobial, anti-mosquito, and spasmolytic properties. Research highlights the antimicrobial potential of resin acid derivatives against bacteria such as *S. aureus* and *E. faecalis*, as well as its anti-mosquito efficacy through insect growth-regulating compounds. The chemical diversity of colophony is influenced by its source and extraction method, with α-pinene being the primary monoterpene in natural varieties. However, synthetic and regional differences contribute to varied chemical profiles. Future research should address the prevalence of ACD, optimize allergen detection, and explore additional therapeutic applications. This review underscores colophony's industrial, pharmacological, and ecological significance while acknowledging its health risks and the need for further studies to mitigate its allergenic potential.

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Introduction

Colophony, or rosin, is made from the resin of coniferous trees, mostly Pinus species. Abietic acid is the most common acidic chemical in it, making up about 90% of the total [1]. Colophony has many uses in industry, including in cosmetics, glues, paints, inks, and soldering fluxes. Rosin improves the soldering process better and cuts down on oxidation when it is applied to a solder joint. Colophony can come from pine tree stumps, tall oil, and oleoresin [2]. Resin acids are the main components of it, but it also has more than a hundred other chemicals in it. Because colophony and its derivatives are cheap, renewable, and good for the environment, the industry has become more interested in them. This is why they are called "green oil." Colophony, which is also known as naval stores, has another benefit: it improves the ship's seals better. Some people may have allergic reactions to both the original and modified forms of rosin, but the modified form may not show up in tests. The main allergens are oxidized resin acids of the abietadiene type, with 15-hydroperoxyabietic acid being the most well-known and still a major sensitizer today [3]. It comes from rosin that has not been altered, but it has been changed in some ways. Some potentially significant positive outcomes may be overlooked when screening with a 20% colophony solution. There is a lot of information out there about colophony's regulatory issues, where it originates, how to diagnose it, how it can be harmful, and how it is used in industry. It is a mix of neutral chemicals and resin acids, such as abietic acid [4]. It is used in many areas, including industry, everyday life, and medical supplies. The fumes that are released during soldering can be dangerous to inhale in if they are heated to the same temperatures as soldering irons. People who are allergic to colophony often experience bronchial asthma that is caused by fumes and contact dermatitis [5]. For a diagnosis of colophony-induced asthma, it may be necessary to perform spirometry and inhalation challenge tests more than once. There are three types of colophony: gum rosin, wood rosin, and tall oil rosin. These types are based on how they are separated from each other. It is found in a wide range of consumer goods [6], both in its original and modified chemical forms. Some of the most common uses for it are as a paper sizing agent, an adhesive, a sealant, and printing ink. Colophony is a well-known allergen and a major cause of ACD because it is used so widely and has strong contact sensitizers like maleopimaric acid in its skin-sensitizing rosin matrix [7]. Further investigation is necessary to ascertain the prevalence of ACD induced by modified rosins. Moreover, such investigations are essential to determine the primary allergenic components responsible for sensitization in individuals with dermatitis and the broader population [8]. Colophony is a resin that comes from the sap of pine trees. A mix of different resin acids and a neutral part that also occurs [9] makes up most of its chemical makeup. Heating colophony releases volatile materials that can cause occupational asthma when inhaled. This information is consistent with the observation that the majority of recorded asthma attacks associated with colophony have occurred among electronics workers utilizing solders containing colophony flux. Exposure to colophony fumes from products like hot-melt glues and hair removal chemicals can also cause occupational asthma [10]. When colophony is heated to more than 180 °C, some of the resin acids break down. People who worked with colophony derivatives, flux-cored solder, or colophony esters in soluble coolant oils have developed asthma. It is important to note that these incidents transpired even when the colophony was heated below this critical threshold [11]. Colophony vapors, which are not uncommon, can also cause contact dermatitis. At first, there was no reversible airway obstruction, but there was coughing and sputum. Colophony-induced occupational asthma was often confused with obstructive bronchitis [12]. Colophony is a solid resin made by distilling the volatile oil from oleoresin obtained from different species of Pinus of Pinaceae. Abietic acid is one of the resin acids that make up colophony. Colophony is also composed of resin, volatile oil, and a bitter-tasting substance. Most of the time, it is used to make ointments and plasters. Colophony is an

important part of veterinary medicine because it is an effective diuretic [13].

Material and Methods

The distillery recieves the crude oleoresin in barrels. It is mixed with water in a heated stainless-steel container, with turpentine comprising about 20% of the weight. After filtering, the mixture is left to sit so that water and other impurities can separate [14]. The diluted oleoresin is then placed into copper or stainless-steel stills, where steam distillation is used to extract the turpentine out. After distillation is complete (which usually takes 1–2 hours for 2.5 tons of oleoresin), the hot resin is poured through strainers into barrels, where it cools down before being prepared for export (Figure 1)[15].

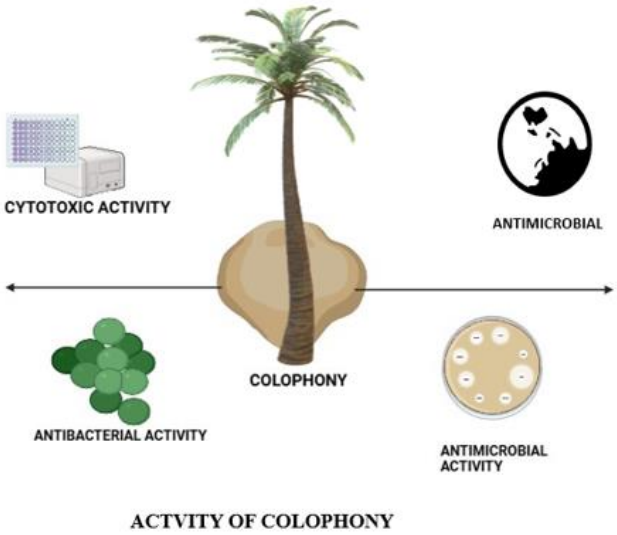


Figure 1: Activity of colophony

Plant Description

Resins represent a class of complex, amorphous compounds. These materials are generally procured from multiple botanical sources, with extraction occurring from plant components like the trunk, bark, root systems, rhizomes, herbaceous plants, seeds, twigs, and similar structures [16]. While acids are employed in certain resin combinations, other methodologies entail the incorporation of resins with either alcohols or esters [17]. Diverse resin categories are obtainable, including oleoresins (constituted by resins and oils), glycoresins (comprising resins and sugars), and oleo-gum-resins (formed from resins, gum, and volatile oils). The botanical origins of such resins are varied, with sources including, but not limited to, asafoetida, ginger, capsicum, myrrh, jalap, and guggul [18]. Many regions of the world cultivate those resins that contain therapeutic herbs. When compared to water, many resins are extremely heavy. In essence, they are transparent or translucent materials [19]. Large carbon atoms are present in them. These goods are either solid or semi-solid. They are typically insoluble in water but soluble in ether, alcohol, benzene, chloral hydrate, fixed oils, and volatile oils. In essence, the resins are metabolites of plants [20]. The resins do not conduct electricity. The resins are initially hard, but when heated or melted, they become soft. Plant injuries from various wounds, such as cuts and incisions, are prevented by the resins. Additionally, the resins shield the plant against infections, worms, insects, and other pests [21]. Resins continue to play a variety of medical roles, including the treatment of eczema, diabetes, ulcers, cancer, fever, arthritis,

constipation, motion sickness, and hypertension. Additionally, some resins are utilized [21].

Chemical Constituents

Analysis of colophony from the Ayvacık (Çanakkale) area revealed that the most abundant monoterpene ingredients included limonene (10.5%), α-pinene (20.5%), linalyl acetate (7.6%), and Δ3-carene (4.7%). With respect to the diterpenoid profile, palustric acid (12.9%), 16-dien-18-oic acid (4.7%), *β*-pinene (5.3%), abietic acid (8.7%), kaur-9(11), and neoabietic acid (1.3%) were the major constituents [22]. The primary monoterpene component of turpentine and all colophonies was α-pinene. However, limonene was the predominant monoterpene in the synthetic colophony. In fact, the percentage of limonene varied throughout each colophony, with the Ayvacık region having the largest amount [23]. In contrast to the profiles of the other colophonies, displayed a poor resin acid profile [24]. Turpentine, as anticipated, was different from all the examined colophonies; it contained a significant concentration of monoterpene hydrocarbons, such as *α*-pinene (30.5%), bornylene (13.5%), limonene (12.3%), and camphene (8.9%), but also included some monoterpene alcohols, esters, and sesquiterpenes (Tables 1, 2).

Table 1: Chemical constituents' percentage of colophony

Name of constituents	Percentage present	[Ref.]
Terpinen-4-ol	1.0	[25]
α-Amorphene	0.1	[26]
α-Cadinol	0.3	[27]
Pregeijerene	1.9	[28]
Cyclododecanone	0.1	[29]
2-Ethyl chromone	0.8	[30]
Ethyl benzoate	0.8	[31]
β-Bourbonene	0.8	[32]
β-Selinene	1.8	[33]
Geijerene	15.4	[34]
(*E*)-Nerolidol	3.0	[35]
Myrcenone	0.1	[36]
Caryophyllene oxide	0.6	[37]
(*E*)-Tsoeugenol	0.1	[38]
α-Humulene	2.2	[38]
δ-Elemene	0.9	[39]
Dihydro-Eudesmol	1.3	[40]
cis-Prenyl limonene	0.3	[41]
Cyperene	0.6	[42]
trans-Nerone	0.9	[41]
trans-β-Terpineol	0.1	[43]
Elemol	3.9	[44]
δ-Cadinene	0.5	[39]
Linalool	3.8	[44]
(*E*)-Caryophyllene	0.6	[45]
Germacrene-D	22.4	[46]
γ-Elemene	6.1	[47]

Table 2: Scientific classification of *P. roxburghii*

Kingdom	Plantae
Division	Pinophyta
Class	Pinopside
Order	Pinales
Family	Pinaceae
Genus	Pinus
Species	*Roxburghii*
Biological name	*P. roxburghii sarg*

Pharmacological Activities

The most common use for colophony is in medicine, where it is found in various plaster-masses and collodions. The stems and roots of different Pinus species are destructively distilled to produce pine or Stockholm tar. It is mostly utilized as an ingredient in bath additives, shampoos, ointments, and creams intended to cure psoriasis and eczema. It is composed of a blend of hydrocarbons and phenols (Figure 2) [48].

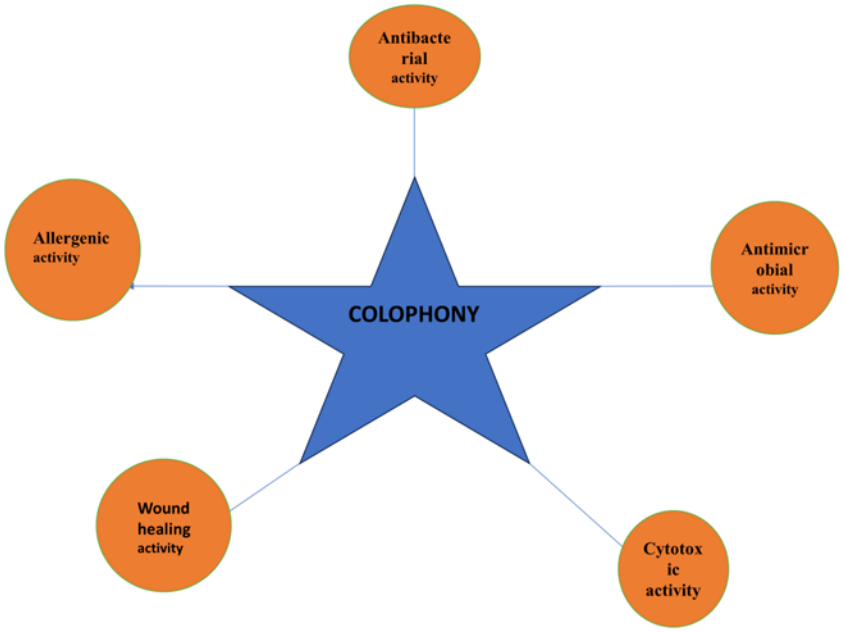


Figure 2: Pharmacological activity

Allergenic

Unmodified colophony is recognized as an etiological factor in contact allergy. The main sensitizing agents are oxidized resin acids, which are formed when the air oxidizes. The first oxidation product from unmodified colophony to be recognized for its contribution to contact allergy was 15-hydroperoxyabietic acid [49]. The isolation and characterization of oxidized resin acids as contact allergens emerged as pivotal themes in subsequent research publications. This body of work also examined the potential for the oxidation of neutral substances to produce allergenic compounds [50]. However, since the neutral fraction makes up only 10% of colophony, any allergens that originate from it are thought to be less important [51]. Changes to chemicals alter how they affect allergies. Hydrogenation and other processes can reduce it, but some changes make it more likely to cause allergies. It is possible to create new allergens, like maleopimaric acid and glyceryl monabietate, that do not react with the

allergens in unmodified colophony. The modified colophony products also contain unmodified material in them, including allergens. This is because the changes are usually halted when the right technical features are reached [52].

Wound healing

The study evaluated the efficacy of *polyalthia longifolia* leaf extracts for wound healing in Wistar albino rats. Both aqueous and methanolic extracts were tested on excision and incision wound models. The results aimed to ascertain the effectiveness of these extracts in facilitating wound repair and regeneration [53]. Both extracts showed a strong ability to help wounds heal, but administering the methanolic extract resulted in faster wound contraction and accelerated epithelization phase [54].

Cytotoxicity

Prinz *et al.* found that the oxidation derivatives of abietic acid and its methyl esters, methyl-7-oxodehydroabietate and methyl-7α-hydroxydehydroabietate, are strong allergens. The KB cells were also used as a model system to test the cytotoxicity of several different compounds. The study found that methyl-7α,13α-dihydroxyabiet-8-14-enoate, methyl-12-oxoabietate, and methyl-7α-hydroxydehydroabietate were the least harmful to cells. Their IC50 values were 12.5, 4.5, and 5.8 µg/mL, which indicates they were not very toxic [55].

Antimicrobial

The function of two resin acid derivatives ws examined from pine rosin, methyl *cis*-7-oxodeisopropyldehydroabietate (MCOD), and methyl trans-7-oxo-deisopropyldehydroabietate (MTOD), that worked as antibacterial agents. A solid-phase assay employing multicell plates was utilized to assess their efficacy against bacterial strains [56]. MCOD and MTOD exhibited antifungal and antibacterial properties by suppressing the proliferation of various filamentous fungi, such as *Mucor racemosus*, *Syncephalastrum racemosum*, Rhizopus stolonifer, and R. arrhizus. They also partially or completely stopped the growth of *Staphylococcus aureus* and *E. faecalis*. When used together, MCOD and MTOD greatly slowed the growth of *E. faecalis*, *S. aureus*, *Klebsiella pneumoniae*, and *Escherichia coli*, showing that they worked better together [57].

Antibacterial

Their antibacterial effects were probably caused by the way their three-dimensional stereo-configuration functioned. It also found that their activity could be improved by synergistic interactions when they are put together. The assay method used in the study was very sensitive and could consistently be repeated for antibacterial and antifungal activity, which made it a good choice for these types of tests [55].

Anti-mosquito

An evaluation was performed on the bioactivity of carbonyl compounds obtained from longifolene (a *polyalthia longifolia* sesquiterpene) against Culex quinquefasciatus mosquitoes, utilizing their oxime ether derivatives. These oxime ethers collectively caused 90–100% mortality, and they exhibited insect growth regulating (IGR) properties at 1 ppm. However, some of these oxime ethers still killed 80–85% of the time, even at a lower dose of 0.1 ppm.

Spasmolytic

Pharmacological investigations demonstrated that abietic acid possesses mild cardiac effects and minimal spasmolytic activity. The therapeutic potential in these domains seems constrained due to the observed weak response [58].

Antioxidant

Metabolic processes or foreign substances produce hydrogen peroxide, superoxide, and hydroxyl radicals [59]. They are found in food, and they quickly react with proteins, lipids, and DNA to harming cells. Antioxidants donate

hydrogen atoms to free radicals. As more people are interested in natural antioxidants, research in various scientific fields has increased [55]. Numerous studies have investigated vitamins, flavonoids, polyphenols, and volatile compounds as antioxidants [58]. Several methods have been developed to evaluate dietary and plant antioxidant capacities. Two methods are recommended by best practices for measuring the antioxidant activity of materials: lipid peroxidation and electron or free radical capture assays, including DPPH, ABTS, ACA, or FRAP, are also recommended [59]. Researchers examined the antioxidant properties of Pinus trees by using their bark, needles, shoots, and seeds. Different solvents and extraction methods made extracts with different chemical compositions. Various approaches were used to test their antioxidant activity, depending on the extracts and their components [60]. The Folin-Ciocalteu reagent exhibited considerable differences in total polyphenol content across various Pinus species and extraction methods. The alcoholic bark extracts of P. radiata (1610 mg GAE/200 mL) and P. brutia (412.42 ± 7.56 mg GAE/g) had much higher levels of polyphenols than the aqueous extracts from P. sylvestris shoots (0.86 ± 0.09 mg GAE/g). This indicates that solvent and plant part significantly influence polyphenol yield. In free radical experiments, both P. brutia bark (9.17 ± 0.13 µg/mL) and P. koraiensis seed (0.023 ± 0.004 mg/mL) methanolic and ethanolic extracts had EC_{50} values that assumed the lowest EC_{50} values and exhibited superior antioxidant activity [61]. A number of studies have shown that mixing water with organic solvents like ethanol, methanol, acetone, isopropanol, or acetonitrile can make plant products more effective at fighting free radicals. These solvent combinations enhance the efficiency of antioxidant component extraction and bioactivity [62]. The antioxidant activity of Pinus densiflora needle extract was evaluated at different concentrations of water and ethanol (0% to 100%). The study aimed to determine the optimal solvent ratio for antioxidant chemical extraction. The extracts of P. densiflora needles with 40% ethanol had the best ability to get rid of radicals, followed by 60%, 20%, 80%, 0%, and 100% ethanol. P. densiflora bark extracts exhibited similar trends, with 40% ethanol demonstrating the highest phenolic content and antioxidant potential [63]. The concentration of the solvent affects the efficacy the extraction process (Table 3).

Table 3: Experimental studies reporting biological activities of Pinus/colophony-derived preparations

Activity	Material	Experimental model	Result	[Ref.]
Antihypertensive	*P. densiflora Sieb. et Zucc.* extract	Normotensive Wistar-Kyoto rats were given tap water, while spontaneously hypertensive rats received tap water, captopril, or Korean red pine bark extract at doses of 50 or 150 $mg.kg^{-1}.day^{-1}$ for seven weeks. Blood pressure was monitored weekly, and following the treatment period, lung, kidney, and serum samples were collected to evaluate ACE activity, angiotensin II levels, and MDA concentrations.	Rats treated with captopril in combination with KRPBE exhibited significantly lower blood pressure compared to the SHR control group. Additionally, both the captopril and KRPBE-treated groups showed markedly reduced ACE activity, angiotensin II levels, and MDA concentrations relative to the SHR control group.	[45]

Hepatoprotective	*P. roxburghii* wood oil	The hepatoprotective effects of *P.* roxburghii wood oil at doses of 200, 300, and 400 mg/kg body weight were evaluated against liver damage induced by ethanol and carbon tetrachloride in rats.	The hepatoprotective effects of *P. roxburghii* wood oil at doses of 200, 300, and 400 mg/kg body weight were evaluated against liver damage induced by ethanol and carbon tetrachloride in rats.	[64]
Antidyslipidemic	*P. roxburghii* needle extracts were fractionated into various solvent-based components, including hexane (B), chloroform (C), n-butanol soluble (D), and n-butanol insoluble (E) fractions. The different fractions were evaluated for their phytochemical profiles and biological activities.	Five solvent fractions (A, B, C, D, and E) derived from *P. roxburghii* needles were administered to six groups of dyslipidaemia hamsters.	In a dyslipidemic hamster model, the extract from *P. roxburghii* needles exhibited a strong ability to reduce plasma lipid profile levels and exerted a beneficial effect on HDL-C and its ratio to total cholesterol.	[60]
Analgesic	*P. roxburghii Sarg*. stem bark ethanolic extract	The analgesic activity was assessed using acetic acid-induced writhing and tail immersion tests in Swiss albino mice.	Mice administered 1% acetic acid exhibited a dose-dependent decrease in abdominal constrictions following treatment with *P. roxburghii Sarg*. alcoholic extract at doses of 100, 300, and 500 mg/kg, with the highest dose (500 mg/kg) achieving 80.95% inhibition of pain.	[65]

Anti-cancer	*P. roxburghii* essential oil	The essential oil was tested for its cytotoxic effects against several human cancer cell lines using the MTT assay. These included HCT-116 (colon cancer), KBM-5 (myelogenous leukemia), U-266 (multiple myeloma), MiaPaCa-2 (pancreatic cancer), A-549 (lung carcinoma), and SCC-4 (squamous cell carcinoma). The study aimed to assess the oil's potential anticancer activity across different cancer types.	It was found that the percentage inhibition of PREO activity was concentration-dependent. The maximum inhibition was 83% for U-266, and 71%, 69%, 73%, 73%, and 76% for HCT-116, SCC-4, MiaPaCa-2, A-549, and KBM-5, respectively.	[66]

Additional Activities of P. Roxburghii

The allergenic potential of 15-hydroperoxydehydroabietic acid (15-HPDA) was evaluated through an experimental sensitization test on guinea pigs, demonstrating its involvement in allergic responses. Nonetheless, *Pinus* species have exhibited antiallergenic properties in various studies. Batta and colleagues documented a clinical case of a 58-year-old woman with a one-year history of lip and perioral dermatitis, presumably associated with exposure to pine-derived substances [67,68]. The patient exhibited a positive allergic response to colophony at both 2 and 4 days during patch testing, and additionally reacted positively to her lipstick after 2 days. It was discovered that colophony existed in the lipstick pigments. After she stopped using the lipstick, her symptoms went away completely, which proved that colophony was the allergen [69].

Health Effects and Allergens of Colophony

Using colophony as a solder flux in soft soldering is a well-known cause of asthma and occupational contact dermatitis. However, there is insufficient detailed information linking specific health outcomes to airborne concentrations of solder fumes. Factors such as soldering temperature and the composition of solder materials, which can affect fume composition, have not been thoroughly studied [70]. Occupational health research has primarily recorded instances of occupational asthma and contact dermatitis associated with exposure to colophony-based products, rather than concentrating on identifying specific allergens in colophony and its derivatives or examining dose-response relationships. It is important to remember that all solder fumes, whether they come from leaded or lead-free solders, can cause health problems like asthma if you breathe them in for a long time. Rosin is a good ingredient in solder wires because it helps the solder flow when it is heated. However, too much exposure to it can cause asthma. To lower these health risks, it is very important to spend money on good ventilation and extraction systems [71]. Colophony or its derivatives have also been found to cause occupational asthma when they are added to products like deodorants, coolant oils, and floor cleaners. People who work in electronics, beauty, poultry processing, welding, adhesives, paper, and tire manufacturing are all at risk. Adding non-colophony solder fluxes has been a step forward, but they might not work for all uses. Even when colophony-free solder is used, taking apart old circuit boards that have colophony on them can still be harmful for health [72].

Asthma at Work

This was one of the first reports of colophony-induced occupational asthma in three welders from different companies. It showed the respiratory risks of being around colophony while soldering [73]. All three welders tested positive for occupational bronchial provocation tests after being exposed to flux-cored solder that contained colophony, which proved that it was the cause of their asthma. Subsequently, Burge and associates executed several studies examining colophony-induced occupational asthma in the electronics sector [74]. In their initial study, Burge *et al.* validated 21 instances of occupational asthma among workers exposed to colophony fumes in the electronics industry, underscoring the considerable health risks linked to colophony exposure in this field. The patients underwent prick tests with prevalent environmental allergens, evaluations of lung function, and provocation tests utilizing colophony-based core solder. All patients experienced bronchoconstriction during the occupational provocation tests utilizing the same solder, and nine exhibited positive responses to one or more prick tests, indicating sensitization. Colophony was determined to be the etiological agent of occupational asthma in these instances. Subsequent investigation by Burge *et al.* revealed a modest yet significant association between atopy, a personal history of allergic disease, and the onset of colophony-induced asthma. Nonetheless, their cross-sectional analysis indicated that smoking and a familial history of allergen sensitivity did not exert a significant influence on the condition [57]. The duration between the initial exposure and subsequent sensitization of the affected workers varied from one to twenty-three years. To determine the prevalence of occupational asthma at a factory producing colophony-cored solder, immunological assessments revealed that 21% of the two most exposed employee groups displayed impairments potentially attributable to asthma. Several studies may have underestimated the prevalence of occupational asthma related to colophony. Colophony-related occupational asthma has occasionally been misidentified as obstructive bronchitis due to symptoms such as coughing and phlegm production, along with the absence of reversible airway obstruction at the initial presentation [75-77]. Colophony is used in soldering flux and a number of other products, such as glue, a mixture for removing chicken feathers at home, and a deodorant made with pine oil in an oil mist

Dermatitis from Direct Contact with the Skin

Colophony is one of the standard allergens used in patch tests for allergies at dermatology clinics in the U.S. and Europe. This helps doctors determine if a patient has a contact allergy to colophony. Epidemiological studies show that about 3.3% of people with colophony sensitivity visit dermatology clinics. Fisher discovered that numerous violinists developed contact dermatitis while using rosin to polish their strings. There have also been cases of contact dermatitis linked to using rosin in paper sizing. Wilkinson and Calnan [78] noted that there have been well-documented cases of dermatitis caused by rosin in anti-slip treatments, such as solutions sprayed on belt surfaces to make them less slippery.

Asthma in the Airways

People who solder electronic parts breathe in colophony vapors that are heated to 350 °C, which is the same temperature as soldering irons. Studies have established a correlation between colophony and bronchial asthma, a condition impacting 4-20% of the population. It is unclear what specific ingredients are in the vapors, but but they are thought to contain resin acids and other substances that irritate the respiratory system as a whole, potentially causing bronchial asthma (~9). Some studies found that workers who soldered experienced long-lasting coughs and breathing problems believed to be caused by allergic alveolitis. This condition was diagnosed using bronchoalveolar lavage, pulmonary function tests, and chest X-rays. Some people who pluck feathers report having asthma attacks. The patients' hypersensitivity tests show that the colophony used to pluck the feathers, not the feathers themselves, is what causes the symptoms. Inhalation challenge testing has demonstrated that employees in tire manufacturing facilities

who handle a glue mixture of pine resin and tall oil are at an increased risk of developing asthma. The histamine inhalation test was administered to both the exposed and control groups to further investigate these findings. The study aimed to ascertain whether occupational exposure to hazardous chemicals correlated with delayed asthma onset and heightened bronchial responsiveness. Results indicated a significant enhancement in responsiveness three hours post-exposure, accompanied by a delayed onset of asthma thereafter. This evidence clearly shows that an inflammatory process that makes the bronchi more responsive, causing delayed asthma [79].

Conclusion

The chemical makeup and overall quality of colophony, a natural resin, depend on the type of pine tree it comes from, how it is made, and how it is stored. Many businesses use its flexibility, including those that make paper sizing, inks, paints, varnishes, dental goods, adhesives, printing inks, and soldering fluxes. Two less common uses are grease removers for shirt collars and powders for removing hair from dead pigs. Humans are exposed to colophony in many physical and chemical forms through a number of different pathways. Colophony has been associated with skin sensitization and occupational asthma; however, the precise allergens responsible for these conditions have not yet identified. Many people believe that oxidized resin acids and some of their derivatives are the main causes of contact dermatitis.

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References

[1] Casal Viqueira, A., Martín, E., Gutiérrez, L., Fonseca, M.E., Navarro, A.T., Valdés, M., Spengler, B., Guerra, S. Oleorresina de pinos. Una nueva fuente para la obtención de aditivos químicos. *CIENCIA Ergo-Sum,* **2015**, 12(1), 64-70.

[2] Llevot, A., Grau, E., Carlotti, S., Grelier, S., Cramail, H. Dimerization of abietic acid for the design of renewable polymers by ADMET. *European Polymer Journal*, **2015**, 67, 409-417.

[3] Sifontes, Á.B., Gutierrez, B., Mónaco, A., Yanez, A., Díaz, Y., Méndez, F.J., Brito, J.L. Preparation of functionalized porous nano-γ-Al_2O_3 powders employing colophony extract. *Biotechnology Reports*, **2014**, 4, 21-29.

[4] Sukarno, A. Physical properties of turpentine and gum rosin Pinus merkusii Jungh et de Vriese tapped oleoresin by borehole method. *The Journal of Experimental Life Science*, **2018**, 8(1), 43-46.

[5] Ferreira, A.G., Fior, C.S., Gualtieri, S.C.J. Oleoresin yield of Pinus elliottii Engelm seedlings. *Brazilian Journal of Plant Physiology*, **2011**, 23, 313-316.

[6] Hasan, A.M., El-Saeed, A.M., Al-Shafey, H.I., El-Sockary, M.A., El-Ghazawy, R.A. A preliminary study on liquid crystalline epoxy curatives from natural abietic acid. *Egyptian Journal of Petroleum*, **2019**, 28(1), 127-136.

[7] Alonso-Esteban, J.I., Carocho, M., Barros, D., Velho, M.V., Heleno, S., Barros, L. Chemical composition and industrial applications of Maritime pine (*Pinus pinaster* Ait.) bark and other non-wood parts. *Reviews in Environmental Science and Bio/Technology*, **2022**, 21(3), 583-633.

[8] Aqil, M., Ali, A., Sultana, Y., Parvez, N. Communication: Matrix type transdermal drug delivery systems of metoprolol tartrate: skin toxicity and *in vivo* characterization. *Ethiopian Pharmaceutical Journal*, **2004**, 22(1), 53-60.

[9] Arrabal, C., Cortijo, M., de Simón, B.F., Vallejo, M.C.G., Cadahía, E. Differentiation among five Spanish Pinus pinaster provenances based on its oleoresin terpenic composition. *Biochemical systematics and Ecology*, **2005**, 33(10), 1007-1016.

[10] Arrabal, C., García-Vallejo, M.C., Cadahia, E., Cortijo, M., de Simón, B.F. Characterization of two chemotypes of *Pinus pinaster* by their terpene and acid patterns in needles. *Plant Systematics and Evolution*, **2012**, 298(2), 511-522.

[11] Sarria-Villa, R.A., Gallo-Corredor, J.A., Benítez-Benítez, R. Characterization and determination of the quality of rosins and turpentines extracted from Pinus oocarpa and Pinus patula resin. *Heliyon*, **2021**, 7(8), 7–17.
[12] Downs, A.M., Sansom, J.E., Colophony allergy: A review. *Contact Dermatitis*, **1999**, 41(6), 305-310.
[13] Beltran, V., Salvadó, N., Butí, S., Cinque, G., Pradell, T. Markers, reactions, and interactions during the aging of pinus resin assessed by Raman spectroscopy. *Journal of Natural Products*, **2017**, 80(4), 854-863.
[14] Bento, A., Escorcio, R., Tome, A.S., Robertson, M., Gaugler, E.C., Malthus, S.J., Pereira, C.S. Pinus radiata bark sequentially processed using $scCO_2$ and an ionic liquid catalyst yields plentiful resin acids and alkanoic acids enriched suberin. *Industrial Crops and Products*, **2022**, 185, 115172.
[15] De Oliveira Junkes, C.F., de Araújo Júnior, A.T., de Lima, J.C., de Costa, F., Füller, T., de Almeida, M.R., Fett-Neto, A.G. Resin tapping transcriptome in adult slash pine (*Pinus elliottii* var. elliottii). *Industrial Crops and Products*, **2019**, 139, 111545.
[16] Lu ChuanWei, L.C., Yu Juan, Y.J., Wang ChunPeng, W.C., Wang JiFu, W.J., Chu FuXiang, C.F. Fabrication of UV-absorbent cellulose-rosin based thermoplastic elastomer via "graft from" ATRP. *Carbohydr Polymer,* **2018**, 188, 128–135.
[17] Osorio Pérez, C.M., Cueva-Gálvez, G.E., Gonzáles Mora, H.E. Caracterización de la oleorresina de Pinus caribaeae obtenido por sistema de pica de corteza con ácido sulfúrico. *Revista Forestal Del Perú*, **2016**, 31(2), 58-68.
[18] Cannac, M., Barboni, T., Ferrat, L., Bighelli, A., Castola, V., Costa, J., Pasqualini, V. Oleoresin flow and chemical composition of Corsican pine (Pinus nigra subsp. laricio) in response to prescribed burnings. *Forest Ecology and Management*, **2009**, 257(4), 1247-1254.
[19] Coppen, J.J.W., Hone, G.A. Gum naval stores: turpentine and rosin from pine resin. *Nonwood Forest Products 2, FAO*, **1995**.
[20] Zhang, D., Zhou, D., Wei, X., Liang, J., Chen, X., Wang, L. Green catalytic conversion of hydrogenated rosin to glycerol esters using subcritical CO_2 in water and the associated kinetics. *The Journal of Supercritical Fluids*, **2017**, 125, 12-21.
[21] Neis, F.A., de Costa, F., de Araújo Jr, A.T., Fett, J.P., Fett-Neto, A.G. Multiple industrial uses of non-wood pine products. *Industrial Crops and Products*, **2019**, 130, 248-258.
[22] Neis, F.A., de Costa, F., de Almeida, M.R., Colling, L.C., de Oliveira Junkes, C.F., Fett, J.P., Fett-Neto, A.G. Resin exudation profile, chemical composition, and secretory canal characterization in contrasting yield phenotypes of Pinus elliottii Engelm. *Industrial Crops and Products*, **2019**, 132, 76-83.
[23] Neis, F.A., de Costa, F., Füller, T.N., de Lima, J.C., da Silva Rodrigues-Corrêa, K.C., Fett, J.P., Fett-Neto, A.G. Biomass yield of resin in adult Pinus elliottii Engelm. trees is differentially regulated by environmental factors and biochemical effectors. *Industrial Crops and Products*, **2018**, 118, 20-25.
[24] Ottavioli, J., Paoli, M., Casanova, J., Tomi, F., Bighelli, A. Identification and quantitative determination of resin acids from Corsican pinus pinaster Aiton oleoresin using 13C-NMR Spectroscopy. *Chemistry & Biodiversity*, **2019**, 16(1), e1800482.
[25] Chen, G.F. Developments in the field of rosin chemistry and its implications in coatings. *Progress in Organic Coatings*, **1992**, 20(2), 139-167.
[26] John, G., Nagarajan, S., Vemula, P.K., Silverman, J.R., Pillai, C. Natural monomers: A mine for functional and sustainable materials–Occurrence, chemical modification and polymerization. *Progress in Polymer Science*, **2019**, 92, 158-209.
[27] Gandini, A. Polymers from renewable resources: A challenge for the future of macromolecular materials. *Macromolecules*, **2008**, 41(24), 9491-9504.
[28] Gaspar, M., Agostinho, B., Fonseca, L., Abrantes, I., de Sousa, H., Braga, M. Impact of the pinewood nematode on naturally-emitted volatiles and $scCO_2$ extracts from Pinus pinaster branches: a comparison with P. pinea. *The Journal of Supercritical Fluids*, **2020**, 159, 104784.
[29] Ghanmi, M., Satrani, B., Aafi, A., Ismail, M.R., Farah, A., and Chaouch, A. Évaluation de la qualité de la colophane du pin maritime (*Pinus pinaster*) et du pin d'Alep (*Pinus halepensis*) du Maroc. *Acta Botanica Gallica*, **2009**, 156(3), 427–435.
[30] Gonçalves, E., Figueiredo, A.C., Barroso, J.G., Henriques, J., Sousa, E., Bonifácio, L. Effect of Monochamus galloprovincialis feeding on Pinus pinaster and Pinus pinea, oleoresin and insect volatiles. *Phytochemistry*, **2020**, 169, 112159.

[31] Wang, H., Malcolm, D.C., Fletcher, A.M. Pinus caribaea in China: introduction, genetic resources and future prospects. *Forest Ecology and Management*, **1999**, 117, 1-15.
[32] Costa, I.L., Alves, A.R., Mulinari, D.R. Surface treatment of Pinus elliottii Fiber and its application in composite materials for reinforcement of polyurethane. *Procedia Engineering*, **2017**, 200, 341-348.
[33] Isikgor, F.H., Becer, C.R. Lignocellulosic biomass: A sustainable platform for the production of bio-based chemicals and polymers. *Polymer Chemistry*, **2015**, 6(25), 4497-4559.
[34] Riveros Alcedo, J.C., Cueva-Gálvez, G.E., Gonzáles Mora, H.E. Evaluación de la oleorresina de Pinus oocarpa en la zona de Oxapampa, Pasco, Perú. *Revista Forestal Del Perú*, **2017**, 32(1), 45-55.
[35] Wang, J., Lu, C., Liu, Y., Wang, C., Chu, F. Preparation and characterization of natural rosin stabilized nanoparticles via miniemulsion polymerization and their pressure-sensitive adhesive applications. *Industrial Crops and Products*, **2018**, 124, 244-253.
[36] Gallo-Corredor, J.A. Sarria-Villa, R.A. Obtencion de colofonia y trementina a partir de la resina de Pinus oocarpa extraida de un bosque industrial en Cauca-Colombia. *Journal de Ciencia e Ingeniería*, 2013, 5(1), 88-91.
[37] Pastor-Bustamante, J.F. Estudio de la calidad de la resina del Pinus caribaea var. caribaea y sus componentes. *Revista Chapingo. Serie ciencias forestales y del ambiente*, **2001**, 7(2), 159-162.
[38] Park, J.Y., Lee, Y.K., Lee, D.S., Yoo, J.E., Shin, M.S., Yamabe, N., Lee, H.J. Abietic acid isolated from pine resin (*Resina Pini*) enhances angiogenesis in HUVECs and accelerates cutaneous wound healing in mice. *Journal of Ethnopharmacology*, **2017**, 203, 279-287.
[39] Kugler, S., Ossowicz, P., Malarczyk-Matusiak, K., Wierzbicka, E. Advances in rosin-based chemicals: the latest recipes, applications and future trends. *Molecules*, **2019**, 24(9), 1651.
[40] Sanap, R., Vikram, S., Sonali, P., Manoj, B., Ishwar, A. A review on role of colophony in different formulation, *International Journal of Scientific Research and Technology*, **2025**, 2(9), 50-58.
[41] Joye Jr, N.M., Lawrence, R.V. Resin acid composition of pine oleoresins. *Journal of Chemical and Engineering Data*, **1967**, 12(2), 279-282.
[42] Mitani, K., Fujioka, M., Uchida, A., Kataoka, H. Analysis of abietic acid and dehydroabietic acid in food samples by in-tube solid-phase microextraction coupled with liquid chromatography–mass spectrometry. *Journal of Chromatography A*, **2007**, 1146(1), 61-66.
[43] Kim, K.H., Daugaard, T.J., Smith, R., Mba-Wright, M., Brown, R.C. Recovery of resin acids from fast pyrolysis of pine. *Journal of Analytical and Applied Pyrolysis*, **2019**, 138, 132-136.
[44] Kopaczyk, J.M., Warguła, J., Jelonek, T. The variability of terpenes in conifers under developmental and environmental stimuli. *Environmental and Experimental Botany*, **2020**, 180, 104197.
[45] Schafer, N., Balwierz, R., Biernat, P., Ochędzan-Siodłak, W., Lipok, J. Natural ingredients of transdermal drug delivery systems as permeation enhancers of active substances through the stratum corneum. *Molecular Pharmaceutics*, **2023**, 20(7), 3278-3297.
[46] Barnola, L.F., Cedeño, A. Inter-population differences in the essential oils of *Pinus caribaea* needles. *Biochemical Systematics and Ecology*, **2000**, 28(10), 923-931.
[47] Berglund, L.A., Burgert, I. Bioinspired wood nanotechnology for functional materials. *Advanced Materials*, **2018**, 30(19), 1704285.
[48] Li, Y., Xu, X., Niu, M., Chen, J., Wen, J., Bian, H., Lai, F. Thermal stability of abietic acid and its oxidation products. *Energy & Fuels*, **2019**, 33(11), 11200-11209.
[49] Mahendra, V.J.A.M., Materials. Rosin product review. *Applied Mechanics and Materials,* **2019**, 890, 77-91.
[50] López-Álvarez, Ó., Zas, R., Marey-Perez, M. Resin tapping: A review of the main factors modulating pine resin yield. *Industrial Crops and Products*, **2023**, 202, 117105.
[51] López-Goldar, X., Lundborg, L., Borg-Karlson, A.K., Zas, R., Sampedro, L. Resin acids as inducible chemical defences of pine seedlings against chewing insects. *PloS One*, **2020**, 15(5), e0232692.
[52] Maiti, S., Ray, S.S., Kundu, A.K. Rosin: A renewable resource for polymers and polymer chemicals. *Progress in Polymer Science*, **1989**, 14(3), 297-338.
[53] Balamuruganvelu, S., Premlal, K.R., Jaikumar, S., Sengottuvelu, S. Wound healing activity of ethanolic

extract of Polyalthia longifolia leaves in excision wound model in rats. *International Journal of Phytotherapy*, **2014**, 78-80.

[54] Lai, M., Zhang, L., Lei, L., Liu, S., Jia, T., Yi, M. Inheritance of resin yield and main resin components in *Pinus elliottii Engelm*. at three locations in southern China. *Industrial Crops and Products*, **2020**, 144, 112065.

[55] Özşen, Ö., Kıran, İ., Dağ, İ., Atlı, Ö., Çiftçi, G. A., Demirci, F. Biotransformation of abietic acid by fungi and biological evaluation of its metabolites. *Process Biochemistry*, **2017**, 52, 130-140.

[56] Yadav, B.K., Gidwani, B., Vyas, A. Polymers, C. Rosin: Recent advances and potential applications in novel drug delivery system. *Journal of Bioactive and Compatible Polymers,* **2016**, 31(2), 111-126.

[57] Burge, P., Harries, M., O'brien, I., Pepy, J.J.C., Allergy, E. Respiratory disease in workers exposed to solder flux fumes containing colophony (pine resin). *Clinical & Experimental Allergy,* **1978**, 8(1), 1-14.

[58] Chauhan, P., Chen, H., Goswami, S.R., Yan, N. Improved mechanical properties of flexible bio-based polymeric materials derived from epoxy mono/di-abietic acid and soyabean oil. *Industrial Crops and Products*, **2019**, 138, 111437.

[59] Shahidi, F., Zhong, Y. Measurement of antioxidant activity. *Journal of Functional Foods*, **2015**, 18, 757-781.

[60] Kim, K.J., Hwang, E.S., Kim, M.J., Park, J.H., Kim, D.O. Antihypertensive effects of polyphenolic extract from Korean red pine (Pinus densiflora Sieb. et Zucc.) bark in spontaneously hypertensive rats. *Antioxidants*, **2020**, 9(4), 333.

[61] Rajčević, N., Nikolić, B., Marin, P.D. Different responses to environmental factors in terpene composition of Pinus heldreichii and P. peuce: ecological and chemotaxonomic considerations. *Archives of Biological Sciences*, **2019**, 71(4), 629-637.

[62] Do, Q.D., Angkawijaya, A.E., Tran-Nguyen, P.L., Huynh, L.H., Soetaredjo, F.E., Ismadji, S., Ju, Y.H. Effect of extraction solvent on total phenol content, total flavonoid content, and antioxidant activity of Limnophila aromatica. *Journal of Food and Drug Analysis*, **2014**, 22(3), 296-302.

[63] Joo, C.G., Lee, K.H., Park, C., Lee, B.C. Antioxidative activities and composition analysis of Pinus densiflora root by ultra high pressure extraction. *Journal of Industrial and Engineering Chemistry*, **2011**, 17(4), 712-716.

[64] Khan, I., Singh, V. Chaudhary, A.K. Hepatoprotective activity of Pinus roxburghii Sarg. wood oil against carbon tetrachloride-and ethanol-induced hepatotoxicity. *Bangladesh Journal of Pharmacology*, **2012**, 7(2), 94-99.

[65] Kaushik, D., Kumar, A., Kaushik, P., Rana, A.C. Analgesic and Anti-Inflammatory Activity of Pinus roxburghii Sarg. *Advances in Pharmacological and Pharmaceutical Sciences*, **2012**, 2012(1), 245431.

[66] Sajid, A., Manzoor, Q., Iqbal, M., Tyagi, A.K., Sarfraz, R.A., Sajid, A. Pinus Roxburghii essential oil anticancer activity and chemical composition evaluation. *EXCLI Journal*, **2018**, 17, 233.

[67] Mirabedini, S., Zareanshahraki, F., Mannari, V. Enhancing thermoplastic road-marking paints performance using sustainable rosin ester. *Progress in Organic Coatings*, **2020**, 139, 105454.

[68] Sacripante, G.G., Zhou, K., Farooque, M. Sustainable polyester resins derived from rosins. *Macromolecules*, **2015**, 48(19), 6876-6881.

[69] Salomé-Abarca, L.F., van der Pas, J., Kim, H.K., van Uffelen, G.A., Klinkhamer, P.G., Choi, Y.H. Metabolic discrimination of pine resins using multiple analytical platforms. *Phytochemistry*, **2018**, 155, 37-44.

[70] Han, W., Liu, F., Li, Y., Liu, G., Li, H., Xu, Y., Sun, S. Advances in Natural Polymer-Based Transdermal Drug Delivery Systems for Tumor Therapy. *Small,* **2023**, 19(35), 2301670.

[71] Silvestre, A.J.D., Gandini, A. Rosin: major sources, properties and applications. In *Monomers, Polymers and Composites from Renewable Resources*, **2008**, 67–88.

[72] Simões, R., Pimentel, C., Ferreira-Dias, S., Miranda, I., Pereira, H. Phytochemical characterization of phloem in maritime pine and stone pine in three sites in Portugal. *Heliyon*, **2021**, 7(4).

[73] Sousa, J.L., Ramos, P.A., Freire, C.S., Silva, A.M., Silvestre, A.J. Chemical composition of lipophilic bark extracts from Pinus pinaster and Pinus pinea cultivated in Portugal. *Applied Sciences*, **2018**, 8(12), 2575.

[74] Sugimoto, N., Kuroyanagi, M., Kato, T., Sato, K., Tada, A., Yamazaki, T., Tanamoto, K. Identification of the Main Constituents in Sandarac Resin, a Natural Gum Base. *Food Hygiene and Safety Science (Shokuhin Eiseigaku Zasshi)*, **2006**, 47(2), 76-79.

[75] Cabaret, T., Gardere, Y., Frances, M., Leroyer, L., Charrier, B. Measuring interactions between rosin and turpentine during the drying process for a better understanding of exudation in maritime pine wood used as outdoor siding. *Industrial Crops and Products*, **2019**, 130, 325-331.

[76] Shingade, G.M., Aamer, Q., Sabale, P., Grampurohit, N., Gadhave, M., Jadhav, S., Gaikwad, D.J. Review on: recent trend on transdermal drug delivery system. *Journal of Drug Delivery and Therapeutics,* **2012**, 2(1), 66-75.

[77] Salvador, V.T., Silva, E.S., Gonçalves, P.G., Cella, R. Biomass transformation: Hydration and isomerization reactions of turpentine oil using ion exchange resins as catalyst. *Sustainable Chemistry and Pharmacy*, **2020**, 15, 100214.

[78] Wang, J., Liu, S., Yu, J., Lu, C., Wang, C., Chu, F. Rosin-derived monomers and their progress in polymer application. *Sustainable Polymers from Biomass,* **2017**, 103-149.

[79] Zinkel, D.F., Russell, J. (Naval stores: Production, chemistry, utilization. *Pulp Chemicals Association*, **1989**.

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